Marawan Eldeib

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Summary

Software engineer and M.Sc. Information Technology student focused on application security and secure system design, with software-engineering experience at Fraunhofer IOSB. Skilled in web application penetration testing and building and hardening full-stack systems. Seeking a full-time position in security or software engineering.

Education

M.Sc. Information Technology (Specialisation: Software Engineering) **Oct 2024 – Mar 2027 (expected)**

University of Stuttgart Stuttgart, Germany

Relevant Coursework:

- **Distributed Systems:** built a fault-tolerant peer-to-peer messaging system in a six-person team with leader election, failure detection, and causal message ordering.
- **IT Service Management & Service Management and Cloud Computing:** cloud computing fundamentals and containerisation (Docker, Kubernetes) alongside IT and cloud service lifecycle management, covering provisioning, automation, and orchestration, with introductory exposure to Ansible, Terraform, and ArgoCD.

B.Eng. (Hons.) Electronics Majoring in Computer **Jul 2020 – Oct 2024**

Multimedia University Cyberjaya, Malaysia

CGPA: 3.7/4.0 — First Class Honours (German equivalent ≈ 1.4).

Work Experience

Student Research Assistant (HiWi) **Oct 2026 – Present**

University of Stuttgart, Institute for Control Engineering of Machine Tools (ISW) Stuttgart, Germany

- **Research software:** supporting the INTERFACE4CPPS research project (ISW with KASTEL at KIT), which develops a software prototype that integrates engineering artifacts and data exchange formats for cyber-physical production systems.

Working Student – Software Engineer **Apr 2026 – Aug 2026**

University of Stuttgart, Institute of Space Systems (IRS), SOFIA Data Center Stuttgart, Germany

- **Performance optimisation:** took a NASA/DLR data-reduction job that could not complete within available memory down to a 14.4 GB peak, without changing the output.
- **Root-cause analysis:** traced the peak to redundant concurrent calculations; a two-line config change gave 39%, and memray-guided dtype fixes a further 32%.

Working Student – Software Engineer **Apr 2025 – Dec 2025**

Fraunhofer IOSB Karlsruhe, Germany

- **LLM evaluation:** tested and evaluated Java and Spring Boot backend services for LLM-based data extraction across Asset Administration Shell (AAS) digital-twin applications, benchmarking six large language models.
- **Test automation:** developed automated end-to-end test suites and curated 30+ evaluation datasets to validate model performance.

Intern – Software Engineer **Jul 2023 – Oct 2023**

Ikhlas Com Sdn Bhd (AirAsia Group) Kuala Lumpur, Malaysia

- **Feature delivery:** resolved 14 defects and shipped 5 checkout-calculation features, improving application stability for 10,000+ users.
- **Refactoring & APIs:** refactored 2 application modules and integrated GraphQL APIs for bilingual content management within Agile sprints.

Projects

Master's Thesis — A GitOps-Constrained Agentic Workflow for System Manipulation **Oct 2026 – Mar 2027**

LLM Agents | GitOps | DevSecOps | HPC | Kubernetes

Supervised by Prof. Dr. Marco Aiello (IAAS), in collaboration with the High Performance Computing Center Stuttgart (HLRS).

Secure GitOps workflow that limits LLM agents to reviewed pull requests on a shared HPC cluster, with sandboxed execution, audit trail and rollback; includes threat modelling and a security evaluation.

Measuring Data Exposure in LLM-Integrated Productivity Tools **Apr 2026 – Sep 2026**

Python | mitmproxy

- **Network traffic analysis:** intercepted outbound HTTPS and WebSocket traffic with Python and mitmproxy, using planted canaries to trace every leaked fragment.
- **Findings:** writing-assistant extensions, Microsoft Word, Edge Copilot and desktop assistants each sent all 12 planted secrets to their servers without any sharing step (no-extension baseline: 0 of 12).

Saha — Secure Full-Stack Platform **Mar 2026 – Present**

Next.js | PostgreSQL | TypeScript

- **Software & security engineering:** designed, built, and deployed a production full-stack booking platform secured with Postgres row-level security, a strict Content Security Policy, MFA-gated admin and audit logging, and conducted a self-audit that uncovered and remediated a privilege-escalation vulnerability.

MangoVision — Aerial Object Detection **Nov 2023 – Jul 2024**

PyTorch | YOLOv8 | Faster R-CNN | Detectron2

- **Deep learning:** developed and benchmarked YOLOv8 and Faster R-CNN detectors across a public benchmark and a custom drone dataset, reaching mAP@0.5 of 0.959 and 0.756. *Best Research Project Award, Gold Medal.*

Publications

M. H. Lye and **M. A. Eldeib**, "Performance Analysis of Faster R-CNN and YOLOv8 Model for Mango Fruits Detection," *Journal of Informatics and Web Engineering*, vol. 5, no. 2, pp. 280–289, 2026. doi: doi.org/10.33093/jiwe.2026.5.2.17

Technical Skills

- **Security:** Web Application Penetration Testing, Burp Suite, OWASP ZAP, OWASP Top 10 (XSS, SQL injection, CSRF, CORS, OAuth 2.0), binary exploitation and exploit mitigation (self-study: buffer overflows, ROP, ASLR, DEP, stack canaries), SIEM and log analysis (Splunk, IBM QRadar).
- **Networks & Protocols:** HTTP/HTTPS, TLS interception, network traffic analysis (mitmproxy, Wireshark), Nmap.
- **Programming:** Python, C/C++, Java, SQL, JavaScript, TypeScript, Flutter.
- **Backend & APIs:** Node.js, Spring Boot, REST APIs, GraphQL, JSON.
- **Web & Frontend:** React, Next.js, HTML, Tailwind CSS.
- **Databases:** PostgreSQL, Redis.
- **Cloud, DevOps & Tooling:** Docker, Git, GitHub Actions, Playwright, AWS (Lambda, API Gateway, DynamoDB, ELB, Auto Scaling), Kubernetes, Terraform, Ansible, ArgoCD, Jenkins.
- **Linux:** Ubuntu, Kali, Bash, YAML, SSH.
- **AI & Data Science:** PyTorch, scikit-learn, MCP.
- **Project Management:** Agile/Scrum, Jira.

Languages: Arabic (Native) | English (C1) | German (B1).